

SQL PROJECT ON SWIGGY DATA



Introduction

A Swiggy SQL project is typically a data analysis case study where you work with a food delivery dataset to answer real-world business questions—like customer behavior, order trends, and revenue drivers. It helps you practice **joins**, **aggregations**, **subqueries**, and **window functions** while simulating how analysts support decision-making in companies like Swiggy.

Dataset Information

The dataset includes information on various restaurants, their locations, cuisines, menus, and more. It is provided in a structured format within the `swiggy.csv` file with around 50,000 rows of data. The schema of the dataset is as follows:

- `restaurant_no`: Unique identifier for each restaurant.
- `restaurant_name`: Name of the restaurant.
- `city`: City where the restaurant is located.
- `address`: Address of the restaurant.
- `rating`: Rating of the restaurant.
- `cost_per_person`: Cost per person for dining at the restaurant.
- `cuisine`: Cuisine offered by the restaurant.
- `restaurant_link`: Link to the restaurant on Swiggy.
- `menu_category`: Category of items on the menu.
- `item`: Name of the menu item.
- `price`: Price of the menu item.
- `veg_or_nonveg`: Indicates whether the item is vegetarian or non-vegetarian.

Project Overview:

- Identifying top-rated restaurants.
- Identifying top city with maximum no. of restaurants.
- Analyzing popular cuisines in different cities.
- Exploring menu categories and items.
- Assessing the cost per person for dining in different places.

Questions:

Basic Level

- How many restaurants have a rating greater than 4.5?
- Which city has the highest number of restaurants?
- How many restaurants have the word "Pizza" in their name?
- What is the most common cuisine among the restaurants in the dataset?
- What is the average rating of restaurants in each city?
- What is the highest-priced item under the "Recommended" menu category for each restaurant?
- Find the top 5 most expensive restaurants that offer cuisines other than Indian.
- Which restaurant provides the lowest average price for all items?

Intermediate Level

- Which restaurant offers the most items in the "Main Course" category?
- Find the restaurants whose average cost is higher than the overall average cost of all restaurants.
- Retrieve the details of restaurants that share the same name but are located in different cities.
- Which top 5 restaurants offer the highest number of categories?

Advanced Level

- List the names of restaurants that are 100% vegetarian, ordered alphabetically.
- Which restaurant provides the highest percentage of non-vegetarian food?
- Determine the most expensive and least expensive cities for dining.
- Calculate the rating rank for each restaurant within its city, where rank = 1.

Solution:

-- Create Database

```
CREATE DATABASE swiggy_db;
```

-- Switch to the database

```
USE swiggy_db;
```

-- Create Tables

```
CREATE TABLE swiggy(  
  restaurant_no INTEGER NOT NULL,  
  restaurant_name VARCHAR(50) NOT NULL,  
  city VARCHAR(10) NOT NULL,  
  address VARCHAR(250),  
  rating DECIMAL(3,1) NOT NULL,  
  cost_per_person INTEGER ,  
  cuisine VARCHAR(50) NOT NULL,  
  restaurant_link VARCHAR(150) NOT NULL,  
  menu_category VARCHAR(100),  
  item VARCHAR(200),  
  price VARCHAR(15) NOT NULL,  
  veg_or_nonveg VARCHAR(10)  
);
```

-- View Tables

```
SELECT * FROM swiggy;
```

-- Import Data into swiggy Table

Basic level:

-- 01 How many restaurants have a rating greater than 4.5?

```
SELECT COUNT(DISTINCT restaurant_no) AS high_rated_restaurants
FROM swiggy
WHERE rating > 4.5;
```

-- 02 Which city has the highest number of restaurants?

```
SELECT city, COUNT(DISTINCT restaurant_name) AS restaurant_count
FROM swiggy
GROUP BY city
ORDER BY restaurant_count DESC
LIMIT 1;
```

-- 03 How many restaurants have the word "pizza" in their name?

```
SELECT COUNT(DISTINCT restaurant_name) AS pizza_restaurants
FROM swiggy
WHERE restaurant_name LIKE '%pizza%';
```

-- 04 What is the most common cuisine among the restaurants?

```
SELECT cuisine, COUNT(cuisine) AS cuisine_count
FROM swiggy
GROUP BY cuisine
ORDER BY cuisine_count DESC
LIMIT 1;
```

-- 05 What is the average rating of restaurants in each city?

```
SELECT city, AVG(rating) AS average_rating
FROM swiggy
GROUP BY city;
```

--06 What is the highest-priced item under the 'recommended' menu category for each restaurant?

```
SELECT restaurant_name, MAX(price) AS highest_price
FROM swiggy
WHERE menu_category = 'recommended'
```

GROUP BY restaurant_name;

-- 07 Find the top 5 most expensive restaurants that offer cuisines other than Indian?

```
SELECT distinct restaurant_name, cost_per_person
FROM swiggy
WHERE cuisine <> 'Indian'
ORDER BY cost_per_person DESC
LIMIT 5;
```

-- 08 Which restaurant provides the lowest average price for all items?

```
SELECT restaurant_name, AVG(cost_per_person) AS avg_price
FROM swiggy
GROUP BY restaurant_name
ORDER BY avg_price
LIMIT 1;
```

Intermediate level:

-- 01 Which restaurant offers the most items in the 'main course' category?

```
SELECT restaurant_name, COUNT(item) AS no_of_items
FROM swiggy
WHERE menu_category = 'Main Course'
GROUP BY restaurant_name
ORDER BY no_of_items DESC
LIMIT 1;
```

--02 Find restaurants whose average cost is higher than the overall average cost?

```
SELECT Distinct restaurant_name, cost_per_person
FROM swiggy
WHERE cost_per_person > (SELECT AVG(cost_per_person) FROM swiggy);
```

-- 03 Retrieve details of restaurants with the same name but located in different cities?

```
SELECT distinct t1.restaurant_name, t1.city, t2.city
FROM swiggy AS t1
JOIN swiggy AS t2
ON t1.restaurant_name = t2.restaurant_name
```

AND t1.city <> t2.city;

-- 04 Which top 5 restaurants offer the highest number of categories?

```
SELECT restaurant_name, COUNT(DISTINCT menu_category) AS no_of_categories
FROM swiggy
GROUP BY restaurant_name
ORDER BY no_of_categories DESC
LIMIT 5;
```

Advance level:

-- 01 List restaurants that are 100% vegetarian, ordered alphabetically?

```
SELECT restaurant_name,
(COUNT(CASE WHEN veg_or_nonveg = 'Veg' THEN 1 END) * 100 / COUNT(*))
AS vegetarian_percentage
FROM swiggy
GROUP BY restaurant_name
HAVING vegetarian_percentage = 100
ORDER BY restaurant_name;
```

--02 Which restaurant provides the highest percentage of non-vegetarian food?

```
SELECT restaurant_name,
(COUNT(CASE WHEN veg_or_nonveg = 'Non-veg' THEN 1 END) * 100 /
COUNT(*)) AS nonvegetarian_percentage
FROM swiggy
GROUP BY restaurant_name
ORDER BY nonvegetarian_percentage DESC
LIMIT 1;
```

-- 03 Determine the most expensive and least expensive cities for dining?

```
WITH city_expense AS (
SELECT city,
MAX(cost_per_person) AS max_cost,
MIN(cost_per_person) AS min_cost
FROM swiggy
```

```
GROUP BY city
)
SELECT city, max_cost, min_cost
FROM city_expense
ORDER BY max_cost DESC;
```

-- 04 Calculate the rating rank for each restaurant within its city (rank = 1)?

```
WITH rating_rank_by_city AS (
    SELECT restaurant_name, city, rating, DENSE_RANK() OVER (PARTITION BY
city ORDER BY rating DESC) AS rating_rank
    FROM swiggy
)
SELECT restaurant_name, city, rating, rating_rank
FROM rating_rank_by_city
WHERE rating_rank = 1;
```